

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281291

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Von Oepen; Randolph

Annuloplasty Ring with Delayed Release

Abstract

An annuloplasty ring system may include an annuloplasty ring. The annuloplasty ring may include a frame having a plurality of cells, the plurality of cells forming a circumferential ring. The annuloplasty ring may also include a skirt coupled to a bottom surface of the frame, and a plurality of biodegradable connectors, each of the plurality of biodegradable connectors connecting two portions of a corresponding one of the plurality of cells. The frame may be formed of a shape-memory material and may be shape-set to a final condition in which the frame has a final diameter. The plurality of biodegradable connectors may maintain the frame in an initial condition having an initial diameter, the initial diameter being larger than the final diameter.

Inventors: Von Oepen; Randolph (Aptos, CA)

Applicant: St. Jude Medical, Cardiology Division, Inc. (St. Paul, MN)

Family ID: 1000008264247

Assignee: St. Jude Medical, Cardiology Division, Inc. (St. Paul, MN)

Appl. No.: 18/951485

Filed: November 18, 2024

Related U.S. Application Data

us-provisional-application US 63563648 20240311

Publication Classification

Int. Cl.: A61F2/24 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to the filing date of U.S. Provisional Patent Application No. 63/563,648, filed Mar. 11, 2024, the disclosure of which is hereby incorporated by reference herein.

BACKGROUND OF THE DISCLOSURE

[0002] Heart valve disease is a significant cause of morbidity and mortality. One treatment for this disease is valve replacement. Native atrioventricular valves (i.e., the tricuspid valve and the mitral valve) typically have a larger size and/or diameter compared to the native aortic valve and the native pulmonary valve. Among the native atrioventricular valves, a regurgitant tricuspid valve typically has a larger size and/or diameter than a regurgitant mitral valve. For example, for patients with severe tricuspid valve regurgitation, the diameter of the tricuspid valve may range from about 40 mm to about 66 mm, although these numbers are merely exemplary.

[0003] Initial human trials of transfemoral replacement of the tricuspid (i.e., right atrioventricular) valve have shown promising results. However, one of the issues associated with right side heart failure (e.g., tricuspid valve regurgitation) is that the annulus of the tricuspid valve is often significantly dilated. As noted above, the tricuspid valve annulus in these patients may have sizes up to **60** mm in diameter or more. If a transcatheter replacement valve is implanted in such a large annulus, it is possible that a significant amount of foreign material (e.g., metal, fabric, or tissue structures of the prosthetic heart valve) will be positioned in a low-flow area near the tricuspid valve. This may create significant risk of blood clotting (e.g., thrombosis), which can result in an embolism (e.g., a pulmonary embolism for the right heart or a stroke for the left heart).

Furthermore the frame of the prosthetic valve may tend to inhibit desirable remodeling of the right ventricle, even if the prosthetic heart valve successfully treats the tricuspid valve regurgitation.

[0004] Another treatment for heart valve disease is annuloplasty, in which the annulus of the heart valve is resized and/or reshaped to help the native valve leaflets more completely coapt to reduce regurgitation or backflow through the native heart valve.

[0005] Annuloplasty procedures are typically performed using a ring or a band that is fixed to the annulus of the native heart valve. In some examples, the annuloplasty ring is a rigid band that is sutured (or otherwise coupled) directly to the native heart valve annulus during an open heart surgery while the patient is on cardiopulmonary bypass. In these types of procedures, the rigidity of the ring forces the native annulus to resize and/or reshape due to the rigidity of the ring. However, open heart surgery is highly invasive and it would be preferable to achieve similar or the same results as surgical annuloplasty rings without needing to perform open heart surgery. Percutaneous annuloplasty rings have been attempted, but have not gained significant traction because the procedures are typically highly complex and very time consuming, with mixed clinical success.

[0006] It would therefore be desirable to be able to treat the tricuspid valve with an annuloplasty procedure to avoid the need for a prosthetic heart valve, while also avoiding the problems of the known annuloplasty procedures.

BRIEF SUMMARY OF THE DISCLOSURE

[0007] According to one aspect of the disclosure, an annuloplasty ring system includes an annuloplasty ring. The annuloplasty ring may include (i) a frame having a plurality of cells, the plurality of cells forming a circumferential ring; (ii) a skirt coupled to a bottom surface of the frame; and (iii) a plurality of biodegradable connectors, each of the plurality of biodegradable

connectors connecting two portions of a corresponding one of the plurality of cells. The frame may be formed of a shape-memory material and may be shape-set to a final condition in which the frame has a final diameter. The plurality of biodegradable connectors may maintain the frame in an initial condition having an initial diameter, the initial diameter being larger than the final diameter. The plurality of cells may each be diamond-shaped. Each of the plurality of diamond-shaped cells may include a top apex, a bottom apex, and two sides apices. Each of the plurality of biodegradable connectors may connect the bottom apex of a corresponding one of the plurality of diamond-shaped cells to the top apex of the corresponding one of the plurality of diamond-shaped cells. One of the side apices of each of the plurality of diamond-shaped cells may be connected to one of the side apices of an adjacent one of the plurality of diamond-shaped cells. The corresponding one of the plurality of diamond-shaped cells may include a top aperture formed in the top apex and a bottom aperture formed in the bottom apex, and the corresponding one of the plurality of biodegradable connectors may have a first end knotted at the top aperture and a second end knotted at the bottom aperture. The plurality of biodegradable connectors may be biodegradable sutures. The frame may be configured to transition from the initial condition to the final condition upon degradation of the biodegradable sutures. The biodegradable sutures may be configured to maintain the frame in the initial condition for more than about one month following implantation of the annuloplasty ring into a patient. The biodegradable sutures may be configured to maintain the frame in the initial condition for more than about two months following implantation of the annuloplasty ring into a patient. The biodegradable sutures may be configured to maintain the frame in the initial condition for about three months following implantation of the annuloplasty ring into a patient.

[0008] Each of the plurality of cells may have an initial height and an initial width in the initial condition of the frame, and a final height and a final width in the final condition of the frame. The initial height may be smaller than the final height, and the final width may be smaller than the initial width. The skirt may be formed of fabric. The skirt may include a first skirt portion and a second skirt portion, the first skirt portion being coupled to the bottom surface of the frame, the second skirt portion being coupled to a top surface of the frame so that the frame is sandwiched between the first skirt portion and the second skirt portion. The skirt may be laminated over the frame to encapsulate the frame therein. The frame may include a plurality of hooks, tines, or barbs extending away from the bottom surface of the frame.

[0009] The annuloplasty ring system may further include adhesives on one or both of (i) the bottom surface of the frame and (ii) the skirt. The annuloplasty ring system may further include a sizing device, and the sizing device may include a plurality of expandable arms. Each of the plurality of expandable arms may have a first end a free end remote from the first end, the first end of each of the plurality of expandable arms being coupled together at a first connection point. The sizing device may include a plurality of actuator arms, each of the actuator arms having a first end coupled to a corresponding one of the plurality of expandable arms, the second end of each of the plurality of actuator arms being coupled together at a second connection point. The sizing device may include a pull wire extending through a coil member, the pull wire being coupled to the second connection point such that pulling the pull wire expands the plurality of actuator arms radially outwardly, and expansion of the plurality of actuator arms radially outwardly causes the plurality of expandable arms to expand radially outwardly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a highly schematic cross-section of a heart illustrating four native heart valves.

[0011] FIG. 2A is a top view of an example of an annuloplasty ring.

[0012] FIG. 2B is a top view of an example of a frame of the annuloplasty ring of FIG. 2A in an

initial assembled state.

[0013] FIG. 2C is a top view of the example of the frame of FIG. 2B in a set shape.

[0014] FIG. 2D is a schematic illustration showing a change in size of a cell of the frame of FIG. 2B over time.

[0015] FIG. 2E is a schematic top view showing a change in configuration of the annuloplasty ring of FIG. 2A over time.

[0016] FIG. 2F is a highly schematic cross-section of the annuloplasty ring of FIG. 2A implanted into a tricuspid valve annulus showing the tricuspid valve annulus changing over time.

[0017] FIG. 3A is a schematic cross-sectional view of a shaping device in contact with the annulus of the native tricuspid valve.

[0018] FIG. 3B is a cut-away top view of the configuration shown in FIG. 3A.

[0019] FIG. 3C is a schematic view of the annuloplasty ring of FIG. 2A having been implanted over the shaping device of FIG. 3A.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0020] This disclosure is generally directed to devices and methods for reducing the size of a tricuspid valve annulus (e.g. tricuspid valve annuloplasty) to reduce or eliminate tricuspid valve regurgitation. As is described in greater detail below, at least some examples described herein rely on an annuloplasty ring or similar device that is implanted into the tricuspid valve annulus in a transcatheter procedure, but which does not reduce the size of the tricuspid valve annulus until after the device is partially or entirely healed (and therefore strongly secured) within the tricuspid valve annulus, which may be a delay of a few weeks or up to three months or more.

[0021] Although the disclosure herein is described in connection with devices and methods to perform tricuspid valve annuloplasty, the concepts described herein may also be applicable to other heart valves, including the mitral valve. However, even if the devices and methods described herein are applicable to both mitral valves and tricuspid valves, they may be particularly suited to the tricuspid valve for a number of reasons. For example, the annulus expansion at a regurgitant tricuspid valve is frequently significantly greater than that found at a regurgitant mitral valve. Further, the right heart is typically less muscular than the left heart, and therefore the tricuspid valve annulus typically requires less force to reduce the size of the annulus compared to the mitral valve. Further, in patients with regurgitant tricuspid valves, the entire right side of the heart can be greatly expanded, and a full replacement valve may not allow remodeling of the right heart. An annuloplasty ring of the type described herein, on the other hand, may continue to reduce the annulus diameter after it has fully healed into the tissue. Still further, patients with tricuspid valve regurgitation are typically more suited to a delayed elimination of regurgitation compared to patients with mitral valve regurgitation, at least in part because mitral valve regurgitation can create more significant symptoms compared to tricuspid valve regurgitation.

[0022] FIG. 1 is a highly schematic cross-section of a heart **10** illustrating four native heart valves, including the tricuspid valve **20**, the mitral valve **30**, the aortic valve **40**, and the pulmonary valve **50**. As can be seen in FIG. 1, the tricuspid valve **20** and the mitral valve **30** are significantly larger than the counterpart aortic valve **40** and pulmonary valve **50**.

[0023] FIG. 2A is a schematic top view of an example of an annuloplasty ring **100a**. The part numbers provided in connection with FIGS. 2A-F ending in an “a” indicate that the component is in an initial assembled state (e.g., just prior to or just after implantation), whereas part numbers ending in a “b” indicate that the component is in a final steady state or a pre-assembly state. Part numbers lacking the “a” or “b” designation are intended to refer to the relevant component generally. In the illustrated example, the annuloplasty ring **100a** may include a frame **120a**, a plurality of connectors **140a**, and a skirt **160a**. In some examples, the frame **120a** may be formed of a shape-memory material, such as a nickel-titanium metal alloy such as nitinol. In some examples, the frame **120a** may be formed by laser cutting a flat sheet of nitinol, although other mechanisms for manufacturing the frame **120a** are suitable.

[0024] FIG. 2B is a top view of the frame **120a** with connectors **140a** assembled to the frame **120a**, but with the skirt **160a** omitted for clarity of illustration. The shape of the assembled frame **120a** include a plurality of diamond-shaped cells attached end-to-end in a continuous circumferential pattern. Although FIG. 2A illustrates a total of twelve diamond-shaped cells, and FIG. 2B illustrates a total of eight diamond-shaped cells, it should be understood that other numbers of cells (including odd numbers of numbers greater than or smaller than eight or twelve) may be suitable. Each diamond-shaped cell may include four struts **122** that form opposing top and bottom apices and opposing side apices. In the illustrated embodiment, the top apex and bottom apex of each cell includes a hole or aperture **124a** formed therein. As will be described in greater detail below, the apertures **124a** may be used to receive connectors **140a**, such as sutures (including biodegradable or bioabsorbable sutures) to temporarily maintain the frame **120a** in the initial assembled condition **120a**. However, in other embodiments, the apertures **124a** may be omitted and instead the connectors may be coupled directly to struts **122a** of the cells (or to other features of the cells).

[0025] FIG. 2C is a top view of the frame **120b** after having been formed and shape-set. The view of FIG. 2C may also correspond to the shape of the frame **120b** after it has been implanted and returned mostly or fully to its set shape. As with FIG. 2B, the skirt is omitted from the view of FIG. 2C for ease of illustration. However, the connectors **140a** are not illustrated in FIG. 2C because, in the configuration of FIG. 2C, the connectors **140a** have either not yet been assembled to the frame **120b** (in order to transition the frame to the configuration of FIG. 2B), or otherwise the connectors **140a** have already degraded and the frame **120a** has transitioned to the condition of frame **120b**. The main difference between frames **120a** and frame **120b** is that the cells of the frame **120b** are in their set-shape in which the top and bottom apices of the cells are farther apart (compared to the condition of frame **120a**) while the side apices of the cells are closer together (compared to the condition of frame **120a**). It should be understood that, in both conditions of frame **120a**, **120b**, the top surface (e.g., the surface visible in FIG. 2A) and the bottom surface (e.g., the surface facing “into the page” in the view of FIG. 2A) may be substantially flat.

[0026] Referring to FIGS. 2A-2C, after forming the frame **120** (e.g., via laser cutting a flat sheet or tube of nitinol), it may be shape-set (e.g., via heat treatment) to have a relatively small outer diameter D_b , as shown in FIG. 2C. After shape-setting the frame **120b** to have the small diameter D_b , it may be temporarily transitioned to a larger diameter frame **120a**. For example, as shown in FIG. 2B, the top and bottom apex of each diamond-shaped cell may be drawn toward each other, which results in a reduction in the height of each cell, but an increase in the circumferential width of each cell, and thus an increase in the diameter D_a of the frame **120a**. While the frame **120a** is in the increased diameter condition, connectors **140a** may be coupled to each cell of the frame **120a** to maintain each cell at the decreased height, despite the fact that the frame **120b** is shape-set to a smaller diameter. For example, a plurality of connectors **140a** may each be tied to a top apex of a cell via one aperture **124a**, and to a bottom apex of the respective cell via another aperture **124a**. The connectors **140a** are preferably secured under tension so that the connectors **140a** prevent the frame **120a** from transitioning back to the set shape in which the height of each cell increases while the width of each cell decreases. Each connector **140a** is preferably formed of as biodegradable suture, thread, or wire that will break down over time within the body following implantation. It should be understood that the diameter D_a of frame **120a** and the diameter D_b of frame **120b** generally refer to the diameter of a circle that passes through the side apices of each cell. However, in other example the relevant diameter may be the outer diameter of the frame or the inner diameter of the frame. In practice, the diameter that may matter the most is the diameter defined by the points of connection of the frame to the annulus tissue, as should become clear below.

[0027] Referring to FIG. 2A, the frame **120a** may be generally annular in the initial condition, and one or more skirts or cuffs **160a** may be coupled to the frame **120a**. In one example, one skirt **160a** is coupled to a bottom surface of the frame **120a** so that, upon implantation, the skirt **160a** will directly contact native tissue of the heart valve annulus (or otherwise tissue adjacent to the annulus,

such as tissue adjacent the annulus facing into the right atrium). The skirt **160a** may be formed of a fabric, or any other material that promotes tissue ingrowth. In some embodiments, a top skirt may also be provided so that the frame **120a** is sandwiched between top and bottom skirts **160a**. In other embodiments, two skirts **160a** may be laminated together over the frame **120a** to encapsulate the frame **120a** therein. Whether one or more skirts **160a** are provided, the skirt(s) **160a** may in some examples be coated or impregnated or otherwise modified with a material (e.g., fibronectin) to promote tissue ingrowth (e.g., by causing a foreign body reaction).

[0028] FIG. 2D is a schematic illustration showing a change in size of a cell of the frame **120** over time. For example the frame **120** may be shape-set so that the struts **122b** of each cell have a relatively large height H_b and relatively small circumferential width W_b (shown on the right of FIG. 2D). When initially forming the annuloplasty ring **100a**, the apices of each cell may be brought together to decrease the height and increase the width, and the connector **140a** may be knotted or otherwise fixed to the apertures **124a** to temporarily maintain the cells in this shorter (e.g., smaller height H_a), wider (e.g., larger width W_a) condition (shown on the left of FIG. 2D). Following implantation, as described in greater detail below, the connector **140a** may degrade over time until it is no longer able to maintain tension between the ends of the connector **140a**, causing the cell to revert (or begin to revert) to its set shape. In other words, during initial assembly, the cells transition opposite the direction of the arrow in FIG. 2D, but over time after implantation, each cell will tend to transition with the direction of the arrow in FIG. 2D after the connector **140a** degrades.

[0029] FIG. 2E is a schematic top view showing a change in configuration of the annuloplasty ring **100a** from its initial condition upon implant to its final configuration after the annuloplasty ring **100b** reverts partly or fully back to its set shape. As can be seen, the annuloplasty ring **100a** may be implanted while it has a temporarily large diameter D_a . Over time, for example from two or three weeks to three or more months, tissue ingrowth will begin to occur on skirt **160a** as the healing process of the patient's body progresses. As the tissue ingrowth occurs, the skirt **160a**, and thus the frame **120a**, become more and more securely fixed to the native valve annulus. Preferably, the connectors **140a** are selected so that they do not significantly degrade until significant tissue ingrowth has occurred and the frame **120a** and skirt **160a** are very securely fixed to the native valve annulus. Once the healing process has completed (or at least progressed significantly), the connectors **140a** may degrade to the point where they can no longer hold tension, and thus can no longer maintain the cells of the frame **120a** in the temporarily wider and shorter condition. As each connector **140a** degrades, the corresponding cell will start to transition back toward its set shape (as shown in FIG. 2D), causing the annuloplasty ring **100b** and frame **120b** to transition back toward the initial set shape with the smaller diameter D_b (as best shown in FIG. 2E). As this transition occurs, the tissue ingrowth has already strongly secured the annuloplasty ring within the native annulus such that the annuloplasty ring **100b** pulls the native annulus to a smaller diameter as it transitions.

[0030] Although not shown in FIGS. 2A-2E, the initial securement of the annuloplasty ring **100a** to the annulus (prior to tissue ingrowth providing enhanced fixation) may be achieved by one or more mechanisms. In some examples, adhesives such as tissue glue may be applied to the bottom surface of the frame **120a** and/or skirt **160a** to assist with the initial fixation. In some examples, including those shown in connection with FIG. 3C below, tines, barbs, or other anchors may be provided on the frame **120a**, including either integrally formed with the frame **120a** or formed separately and later connected. In further examples, a secondary procedure may be performed to drive separate anchors (e.g., barbs, "T"-shaped anchors, corkscrew anchors, etc.) through the frame **120a** and into the tissue immediately after placing the annuloplasty ring **100a** in contact with the tissue. It should be understood that one or more of these options, or other suitable options for initial anchoring, may be combined.

[0031] FIG. 2F is a highly schematic illustration showing annuloplasty ring **100a** implanted within

the annulus of the native tricuspid valve **20a**. In FIG. 2F, for simplicity of illustration, the annuloplasty ring **100a**, **100b** is shown only as a dashed circle. As can be seen in FIG. 2F, the top of the figure represents an initial condition in which the annuloplasty ring **100a** has the temporarily large diameter, and the native tricuspid valve **20a** has a relatively large diameter and the leaflets of the tricuspid valve are not efficiently closing. Over time, after the annuloplasty ring has been “locked” into the annulus via tissue ingrowth, the connectors **140a** degrade to allow the annuloplasty ring **100b** to transition back to its smaller diameter size. As shown in the bottom view of FIG. 2F, after the annuloplasty ring **100b** has transitioned partially or fully back to its original set shape, it pulls the annulus of the tricuspid valve **20b** inwardly to reduce its size, enhancing the coaptation of the leaflets of the tricuspid valve **20b**, reducing or eliminating regurgitation.

[0032] In some examples, the connectors **140a** may be selected to degrade over a desired timepoint, for example after about 2 or 3 weeks, after about 1 month, after about 2 months, or about after 3 months. The degradation time may be fine tuned in some examples by the type and the size of the connector **140a**. In some examples, the connectors **140a** may be formed of biodegradable polymers, including for example polyglycolic acid (“PGA”), polylactic acid (“PLA”), polycaprolactone (“PCL”), or copolymers thereof. In some examples, each connector **140a** may be substantially identical to each other connector. However, in other examples, different connectors **140a** may be used to stagger the amount of time the connectors take to degrade. For example, some connectors **140a** may be configured to degrade after 4 weeks, some connectors **140a** may be configured to degrade after 6 weeks, some connectors **140a** may be configured to degrade after 8 weeks, some connectors **140a** may be configured to degrade after 10 weeks, and some connectors **140a** may be configured to degrade after 12 weeks. With this type of configuration, individual cells of the frame **120a** will begin to transition back to their set shape in a staggered timeline, which may allow for a more gradual decrease in diameter of the frame **120b**, compared to all of the connectors **140a** degrading at about the same time which may result in a large amount of force over a small amount of time. It should be understood that the particular combination of connectors **140a** (if different connectors are used) may be any suitable combination to allow for a more gradual diameter shift, whether there are only two different types of connectors **140a** or more than two different types of connectors **140a**.

[0033] As should be understood from the above description, one of the goals of the annuloplasty ring **100** is to force the tricuspid valve annulus back into or near its original shape prior to its dilation as part of the heart disease process. Because the expansion or dilation of the tricuspid valve annulus typically does not occur over a very short period, it may be beneficial to have a correspondingly slow reduction in the size of the tricuspid valve annulus via the transition of the annuloplasty ring from the initial condition **100a** back toward the shape-set condition **100b**. Nitinol is one desirable option for forming frame **120** because, after the connectors **140a** degrade, the nitinol frame **120** will continuously try to reach its original set shape. The frame **120** of annuloplasty ring **100** will at all times be in an equilibrium of forces after implantation. When all the connectors **140a** have degraded, the frame **120a** will reduce its diameter until an equilibrium of force has been achieved (which accounts for opposing forces applied by the native tissue on the annuloplasty ring). However, if the frame **120a** has not yet achieved its final diameter D_b , it will continue to shrink over time. Nitinol is also flexible, and a flexible frame **120** may be advantageous as the heart muscle is in constant motion and the flexibility of the frame **120** will not unduly hinder necessary motion of the heart muscle, particularly compared to stiff, surgically implanted annuloplasty rings or surgically implanted prosthetic heart valves.

[0034] Although the illustrations of FIG. 1 and FIG. 7 illustrate the tricuspid valve **20** as a circle, in practice, the annulus of the tricuspid valve **20**, particularly in patients that have a disease state necessitating treatment, is typically not a circle. Because the annuloplasty ring **100a** may form a circle, a mismatch in shape between the annuloplasty ring **100a** and the annulus of the tricuspid valve **20** may exist prior to performing the annuloplasty procedure. This may be problematic,

because in order for tissue ingrowth to secure the annuloplasty ring **100a** to the tissue, contact between the tissue and the annuloplasty ring **100a** is typically needed. In other words, tissue ingrowth will not occur, or at least is less likely to occur, at locations where the annuloplasty ring **100a** is not in contact with tissue. Further, because the particular size and shape of the annulus will not be identical among all patients, a single shape of annuloplasty ring **100a** will not likely be achieve good contact with the tissue of all patients without an additional feature.

[0035] FIG. **3A** is a schematic cross-sectional view of a shaping device **200** in contact with the annulus **22** of the native tricuspid valve **20**. The shaping device **200** may include a plurality of expanding arms **210**. The expanding arms **210** may each extend from a first end to a second free end. The first ends of each of the expanding arms **210** may come together (e.g., be connected) at a central location, such as a ring or connector **211**, and the free ends of each expanding arm may include contours, such as hooks or ramped extensions, configured to hook around the annulus **22**. In the illustrated example, each free end of each expanding arm **210** may include a straight segment **212** configured to press against an inner surface of the annulus **22**, and an angled terminal segment **214** configured to hook around the bottom or outflow end of the annulus. In the illustrated embodiment, a total of twelve expanding arms **210** (shown in FIG. **3B**) are included at substantially equal intervals around the circumference of the shaping device **200**. However, in other embodiments, more or fewer expanding arms **210** may be provided. In some embodiments, at least three expanding arms **210** are included to assist with self-centering of the shaping device **200**. Further, in some embodiments, the expanding arms **210** (if formed of shape-memory material such as nitinol) may be shape set (e.g., via heat treatment) to the expanded condition similar to that shown in FIG. **3A**.

[0036] Still referring to FIG. **3A**, an actuator arm **216** may be attached to each expanding arm **210**. In some embodiments, the actuator arms **216** are integral with the expanding arms **210**, and may be attached via a hinge such as a living hinge. In other embodiments, the actuator arms **216** are formed separately and attached to the expanding arms **210**. Each actuator arm **216** may include a first end coupled to a corresponding expanding arm **210**, and a second end attached to the other actuator arms **216**. With this configuration, while the expanding arms **210** generally extend radially outward from a central longitudinal axis (when in the expanded condition), the actuator arms **216** generally extend from the extending arms **210** toward the central longitudinal axis. At the point of connection between the plurality of actuator arms **216**, a ring **218** or similar structure may be provided. In some examples, the expanding arms **210** and actuator arms **216** are formed from the same material, such as stainless steel, nitinol, or other metals or metal alloys. In other examples, the expanding arms **210** and actuator arms **216** may be formed of different materials.

[0037] In some embodiments, an actuator or pull wire **220** may be provided as part of the shaping device **200**. In some examples, the pull wire **220** may be formed of a cable, which may be made from metal or metal alloys, including stainless steel. The pull wire **220** may have a terminal end that is coupled to the ring **218**, or in some examples may be looped through the ring **218**. In either example, an end of the pull wire **220** is preferably accessible outside the body by a user so that the pull wire **220** may be manually pulled (or pulled via an actuator on a handle of a delivery device that is connected to the pull wire **220**). In use, as the shaping device **220** is delivered into or near the annulus **22**, the expanding arms **210** and actuator arms **216** may extend parallel or near parallel to each other and to a central longitudinal axis of the shaping device **200** and/or annulus **22**. With the expanding arms **210** positioned interior to the annulus **22**, the user may retract pull wire **220**. Proximal force on pull wire **220** is transmitted to ring **218**, tending to pull ring proximally, causing the actuator arms **216** to “open” away from each other and thus causing the expanding arms **210** to “open” away from each other. As the expanding arms **210** move radially outwardly, they eventually contact the inner surface of the annulus **22**, for example with the straight segments **212** directly contacting the annulus **22** and the angled terminal segments **214** hooking around the outflow portion of the annulus **22**, helping to keep the expanding arms **210** in contact with the annulus **22**.

As additional proximally-directed force is applied to the pull wire **220**, the expanding arms **210** will continue to expand, forcing the annulus **22** to reshape into a generally circular shape, as best shown in FIG. **3B**. Further, the expanding arms **210** may form a general cone shape at this stage, with the smaller diameter end of the cone shape positioned above the annulus **22** and the larger diameter of the cone shape positioned at or adjacent to the annulus **22**. However, even if the annulus **22** is not reshaped to a perfect circle, as will become clear below, the shaping device **200** may be used to deliver and/or deploy the annuloplasty ring **100a** (e.g., using the ramped surface defined by the cone shape of the expanding arms **210**), such that both the annuloplasty ring **100a** and the annulus **22** take the same shape defined by the shaping device **200**.

[0038] It should be understood that the shaping device **200** may be delivered to the patient through a transvascular catheter (e.g., a catheter extending in the antegrade direction thorough the femoral vein, through the inferior vena cava, into the right atrium, and extending into or near the tricuspid valve) while the expanding arms **210** and actuator arms **216** are in a collapsed condition. In some examples, a coil **230** may be coupled to and/or in contact with the ring or connector **211** and/or the connected ends of the expanding arms **210**. In this example, the pull wire **220** may extend through a central axis defined by coil **230**. If the coil **230** is included, when the pull wire **220** is pulled proximally, the coil **230** may help provide a counterforce so that the expanding arms **210** and actuator arms **216** expand outwardly instead of simply also pulling proximally.

[0039] FIG. **3C** is a schematic view of the annuloplasty ring **100a** having been implanted over shaping device **200** and onto the annulus **22**. In an exemplary method of use, a delivery catheter **300** may be advanced into the right atrium, which may for example be a steerable delivery catheter **300**. As the delivery catheter **300** is delivered to or adjacent to the annulus **22**, the shaping device **200** may be in a collapsed condition within the delivery catheter **300**. In other examples, the delivery catheter **300** may be delivered first, and then the shaping device **200** may be advanced through the delivery catheter **300**. The shaping device **200** may be pushed distally out of the delivery catheter **300** (or the delivery catheter **300** may be withdrawn relative to the shaping device **200**) so that the shaping device **200** exits the catheter **300**. Upon exiting the catheter **300**, the shaping device **200** may begin to expand. In some examples, the expansion may be at least in part due to the expanding arms **210** tending to self-expand. In some examples, the expanding arms **210** may be forced to open by pulling pull wire **220** proximally. In some examples, the expansion may be due to both. Regardless of the particular method to achieve expansion, the expansion arms **210** may be expanded into contact with the interior surface of the annulus **22** (e.g., as shown in FIGS. **3A-3C**), and further expanded to force the annulus **22** into a generally circular shape (although in some embodiments this further expansion is not necessary, or results in a non-circular shape of the annulus **22**). The steps described above in some examples will result in a configuration similar to that shown in FIGS. **2A-2B**. Once that configuration (or a similar configuration is achieved), the annuloplasty ring **100a** may be delivered and deployed.

[0040] Referring to FIG. **3C**, with the annuloplasty ring **100a** in the initial assembled state, it may be advanced through catheter **300** (e.g., while it is in a partially or fully collapsed condition. The annuloplasty ring **100a** may be advanced by a pusher device in some examples. In some examples, the center opening of the annuloplasty ring **100a** may be positioned over the pull wire **220** and/or coil **230**, with the pull wire **220** and/or coil **230** acting as a rail over which the annuloplasty ring **100a** can be delivered. As noted above, the frame **120a** may include anchoring elements, which in FIG. **3C** are depicted as barbs **128a**, although other anchoring elements may be suitable. If barbs **128a** are included, they may be integral with the frame **120a**, or formed separately and attached to the frame **120a**. During the delivery, if barbs **128a** (or another fixation mechanism such as adhesives, hooks, or tines are positioned on the frame **120a** and/or the skirt **160a**), the annuloplasty ring **100a** is preferably advanced over the pull wire **220** and/or coil **230** with the barbs **128a** (or other fixation mechanism) facing toward the annulus **22** (e.g., facing the distal or leading end of the catheter **300**). Furthermore, in some examples, a capture wire **180** may be coupled to the

annuloplasty ring **100a** during the advancing. In some examples, the capture wire **180** may be a suture or similar wire that is releasably coupled to the annuloplasty ring **100a**. In some example the annuloplasty ring **100a** may include a ring **190** or similar structure that the capture wire **180** may be coupled to, although in other embodiments the capture wire **180** may be connected to a strut of the frame of the annuloplasty ring **100a**. In some examples, the capture wire **180** may be a suture that is looped through ring **190**, with both ends of the suture accessible outside the patient. While the capture wire **180** is coupled to the annuloplasty ring **100a**, it may be retrievable back into the catheter **300** by pulling on the capture wire **180**. When it is desired to decouple the annuloplasty ring **100a** from the capture wire **180**, one end of the capture wire may be pulled, allowing the other end to travel toward the ring **190**, through the ring **190**, and then back through the catheter **300** to fully decouple the capture wire **180** from the annuloplasty ring **100a**.

[0041] Whether or not capture wire **180** is included, during delivery the pusher device may be advanced distally to push the annuloplasty ring **100a** out of the open distal end of the catheter **300**, and over the expanding arms **210**. As described above, the expanding arms **210** may form a general conical and/or ramped surface over which the annuloplasty ring **100a** may be forced. As the annuloplasty ring **100a** slides over the expanding arms **210**, due at least in part to the flexible nature of the frame **120a**, the annuloplasty ring **100a** will generally conform to the shape of the expanding arms **210**, which matches the shape of the annulus **22** since the annulus **22** has been reshaped by the expanding arms **210**. The annuloplasty ring **100a** may be further pushed down the ramp formed by the expanding arms **210** until the barbs **128a** (or other fixation mechanism(s)) contact and fix into the tissue of the annulus **22**, as shown in FIG. 3C. At this point, the sealing skirt **160a** (or a portion thereof) is sandwiched between the frame **120a** and the annulus **22**, and the barbs **128a** (or other fixation mechanisms) provide for an initial fixation so that the annuloplasty ring **100a** stays coupled to and in contact with the annulus **22**. At this point, the capture wire **180** (if included) may be withdrawn, disconnecting the annuloplasty ring **100a** from the catheter **300**. Then, any applied force on the pull wire **220** may be released and the expanding arms **210** may be re-collapsed into the catheter **300**, for example by advancing the catheter **300** distally to force the expanding arms **210** to re-collapse. Then, the catheter **300** may be withdrawn with the sizing device **200** (which may also be referred to as a shaping device) inside the catheter **300**, completing this stage of the procedure.

[0042] As should be understood from the above, after this stage of the process is completed, the patient's heart may have the configuration of the top view of FIG. 2F, and the annuloplasty ring **100a** may have the configuration in the top view of FIG. 2E. Further active intervention by the user is no longer needed at this point. Rather, tissue ingrowth will naturally occur to strongly fix the annuloplasty ring **100a** to the patient's annulus **22** over the following days and weeks. The connectors **140a** will similarly naturally begin to degrade without further active intervention, but preferably not until significant tissue ingrowth has occurred. As noted above, the connectors **140a** may be selected to degrade only after a desired time has passed (e.g., three months), and in some examples different connectors **140a** may be used to provide for staggered degradation of the different connectors **140a** over time. As the connectors **140a** degrade and the individual cells start to transition as shown in FIG. 2D, the annuloplasty ring **100a** starts to transition to the bottom configuration of annuloplasty ring **100b** shown in FIG. 2E, drawing the annulus **22** to a smaller diameter as shown in the transition of FIG. 2F, ultimately enhancing coaptation and reducing regurgitation across the native tricuspid valve **20b**.

[0043] It should be understood that the shaping device **200** shown and described above may be particularly suited to situations in which the annuloplasty ring **100a** has a mismatch in shape compared to the annulus being treated. If the annuloplasty ring **100a** is not designed specifically for a particular patient, given the large patient-to-patient variety in anatomy (particularly at a diseased tricuspid valve), the annuloplasty ring **100a** will typically not match the patient's anatomy. As should be understood from the above, this mismatch may be overcome via the use of the shaping

device **200**. However, in other embodiments, the annuloplasty ring **100a** may be provided as a patient-specific device. For example, prior to performing the annuloplasty procedure, the patient's anatomy may be imaged, for example using CT imaging or using another imaging modality. The shape of the particular patient's anatomy (e.g., the shape of the tricuspid valve annulus) may be determined from the imaging, and the annuloplasty ring **100a**, and particularly the frame **120a**, may be shape-set to match the patient anatomy. If the annuloplasty ring **100a** is personalized to the patient, in some examples the shaping device **200** may be omitted, such that the annuloplasty ring **100a** may be directly pressed into contact with the annulus **22**, maintaining good contact to allow tissue ingrowth due to the matching shape between the annuloplasty ring **100a** and the annulus **22**. [0044] Further, although addressed above, it is reiterated here that the concepts relating to annuloplasty ring **100a** described here may be used with annuloplasty rings to treat the mitral valve, aortic valve, or pulmonary valve, even if the annuloplasty ring **100a** is particularly well suited to treat the tricuspid valve.

[0045] Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. An annuloplasty ring system comprising: an annuloplasty ring, comprising: a frame having a plurality of cells, the plurality of cells forming a circumferential ring; a skirt coupled to a bottom surface of the frame; and a plurality of biodegradable connectors, each of the plurality of biodegradable connectors connecting two portions of a corresponding one of the plurality of cells, wherein the frame is formed of a shape-memory material and is shape-set to a final condition in which the frame has a final diameter, the plurality of biodegradable connectors maintaining the frame in an initial condition having an initial diameter, the initial diameter being larger than the final diameter.
2. The annuloplasty ring system of claim 1, wherein the plurality of cells are each diamond-shaped.
3. The annuloplasty ring system of claim 2, wherein each of the plurality of diamond-shaped cells includes a top apex, a bottom apex, and two sides apices.
4. The annuloplasty ring system of claim 3, wherein each of the plurality of biodegradable connectors connect the bottom apex of a corresponding one of the plurality of diamond-shaped cells to the top apex of the corresponding one of the plurality of diamond-shaped cells.
5. The annuloplasty ring system of claim 4, wherein one of the side apices of each of the plurality of diamond-shaped cells is connected to one of the side apices of an adjacent one of the plurality of diamond-shaped cells.
6. The annuloplasty ring system of claim 4, wherein the corresponding one of the plurality of diamond-shaped cells includes a top aperture formed in the top apex and a bottom aperture formed in the bottom apex, the corresponding one of the plurality of biodegradable connectors having a first end knotted at the top aperture and a second end knotted at the bottom aperture.
7. The annuloplasty ring system of claim 4, wherein the plurality of biodegradable connectors are biodegradable sutures.
8. The annuloplasty ring system of claim 7, wherein the frame is configured to transition from the initial condition to the final condition upon degradation of the biodegradable sutures.
9. The annuloplasty ring system of claim 8, wherein the biodegradable sutures are configured to maintain the frame in the initial condition for more than about one month following implantation of the annuloplasty ring into a patient.
10. The annuloplasty ring system of claim 8, wherein the biodegradable sutures are configured to

maintain the frame in the initial condition for more than about two months following implantation of the annuloplasty ring into a patient.

11. The annuloplasty ring system of claim 8, wherein the biodegradable sutures are configured to maintain the frame in the initial condition for about three months following implantation of the annuloplasty ring into a patient.

12. The annuloplasty ring system of claim 1, wherein each of the plurality of cells has an initial height and an initial width in the initial condition of the frame, and a final height and a final width in the final condition of the frame, the initial height being smaller than the final height, the final width being smaller than the initial width.

13. The annuloplasty ring system of claim 1, wherein the skirt is formed of fabric.

14. The annuloplasty ring system of claim 1, wherein the skirt includes a first skirt portion and a second skirt portion, the first skirt portion being coupled to the bottom surface of the frame, the second skirt portion being coupled to a top surface of the frame so that the frame is sandwiched between the first skirt portion and the second skirt portion.

15. The annuloplasty ring system of claim 1, wherein the skirt is laminated over the frame to encapsulate the frame therein.

16. The annuloplasty ring system of claim 1, wherein the frame includes a plurality of hooks, tines, or barbs extending away from the bottom surface of the frame.

17. The annuloplasty ring system of claim 1, further comprising adhesives on one or both of (i) the bottom surface of the frame and (ii) the skirt.

18. The annuloplasty ring system of claim 1, further comprising: a sizing device comprising: a plurality of expandable arms, each of the plurality of expandable arms having a first end a free end remote from the first end, the first end of each of the plurality of expandable arms being coupled together at a first connection point.

19. The annuloplasty ring system of claim 18, wherein the sizing device includes a plurality of actuator arms, each of the actuator arms having a first end coupled to a corresponding one of the plurality of expandable arms, the second end of each of the plurality of actuator arms being coupled together at a second connection point.

20. The annuloplasty ring system of claim 19, wherein the sizing device includes a pull wire extending through a coil member, the pull wire being coupled to the second connection point such that pulling the pull wire expands the plurality of actuator arms radially outwardly, and expansion of the plurality of actuator arms radially outwardly causes the plurality of expandable arms to expand radially outwardly.
